

Virfie: Virtual Group Selfie Station for Remote Togetherness

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ABSTRACT

Selfies have become a prominent means of online communication. Group selfies, in particular, encourage people to represent their identity as part of the group and foster a sense of belonging. During the COVID-19 pandemic, video conferencing systems are used as a tool for group selfies. However, conventional systems are not ideal for group selfies due to the rigidity of grid-based layout, information overload, and lack of eye contact. To explore design opportunities and needs for a novel virtual group selfie platform, we conducted a participatory design, and identified three characteristics of virtual group selfie scenarios, “context with narratives”, “interactive group tasks”, and “capturing subtle moments.” We implemented Virfie, a web-based platform that enables users to take group selfies with embodied social interaction, and to create and customize selfie scenarios using a novel JSON specification. In order to validate our design concept and identify usability issues we conducted a user study. Feedbacks from the study participants suggest that Virfie is effective at strengthening social interaction and remote togetherness.

CCS CONCEPTS

• Human-centered computing → Interactive systems and tools.

KEYWORDS

Virtual group selfie, Video conferencing, Social bonds, Participatory design, Remote togetherness

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1 INTRODUCTION AND BACKGROUND

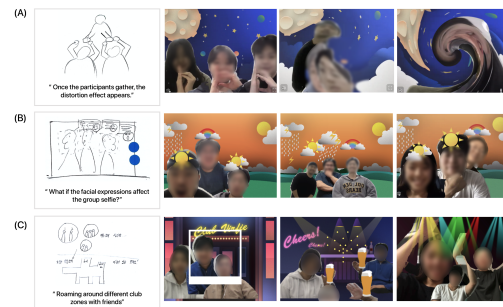


Figure 1: Idea sketches (left-most) and group selfies of three scenes in remote situations. (A) Space Wiggle: When users gather at the center, the vortex distortion effect applies. (B) Emotion Tracker: Weather icons above users represent emotional states. (C) Club Virfie: Users explore lobby, bar, and dance floor of Club Virfie.

Sharing photographs and videos online through social media services has become a common activity, particularly among social networking sites and messaging apps [2, 11]. According to Kelsey [6], “We are moving away from photography as a way of recording and storing the past, and instead of turning photography into a social medium in its own right,” Selfies have become a global phenomenon that is pervasive on social media networks as well as an emerging personal media genre [3, 5, 15, 17–19]. Svelander [16] proposes that, selfies are not only for capturing self-images but also facilitate interactions between individuals by sharing captured images on social media. Particularly, a group selfie is digital self-portrait of an individual, which encourages social communications as well as provide opportunities to represent one’s own identity as part of a group and share with groups of varying sizes [3, 11].

As the COVID-19 social distancing order restricts in-person social events, meetings, and entertainment [14], most group selfies are performed on a video conferencing system such as Zoom, Skype,

and Google Meet. Despite the widespread use of such systems, researchers have identified various limitations of conventional video conferencing as a tool for building social bonds. For example, the grid views are not suitable for users to feel cohesion with other people in the group [7] or to have spatial awareness [9]. Video fatigue [12] is another common issue caused by the lack of direct eye gaze [13], information overload as they try to focus on multiple faces at the same time [1], and all in one-inch boxes that often jump from one position on the screen to another as different individuals speak simultaneously [9].

To address these difficulties related to online group selfies, several products have attempted to adjust the video layout during a virtual meeting. Microsoft¹ and Skype² put meeting participants together against a shared digital background in together mode. Mmhmm³ recently updated their interactive group calls where participants are out-of-the-box and have multiplayer chats in virtual settings. Apple's Facetime⁴ floats participants' grid around the screen and resizes it according to the speaker. Moreover, Bunch⁵ provides different sizes of grids to create a way for participants to party up with friends in various game themes.

While several products mentioned above have started using alternative layouts other than grids, tools for group selfies have much room for improvements. For instance, it is critical to understand what form of content drives engagement and social bonds. Also, there are few studies that examined user needs, required interactivity, and missing functionalities for supporting virtual group selfie activities. Given the situation, finding what users want while taking a virtual group selfie would be an important research goal.

To examine design opportunities for a novel virtual group selfie station, we first conduct a participatory design. By analyzing proposed scenarios of virtual group selfie experience, we identify three characteristics of virtual group selfies: context with narratives, interactive group tasks, and capturing a subtle moment. We then present Virfie, a virtual group selfie station, and three sample scenarios as illustrated in Figure 1. Finally, we report a user study to investigate how users perform and take group selfies remotely with friends.

The contributions of this paper are threefold: 1) With the participatory design session, we explore the possibilities of online group selfie ideas. 2) To allow non-programmers to create and modify a wide range of scenarios, we developed a JSON specification grammar and a built-in editor. 3) Through a user study, we validate if the Virfie supports the design goals and identify usability issues, unmet needs, or further opportunities. We believe Virfie and the user study together demonstrate the potential value of virtual group selfie systems for promoting social bonds. Findings presented in this paper would be helpful for designers and developers who want to strengthen social bonds of people in remote situations.

¹Microsoft Teams - Video Conferencing, Meetings, Calling. Retrieved January 12, 2022 from <https://www.microsoft.com/en-us/microsoft-teams/group-chat-software>

²Skype - Stay connected with free video calls worldwide. Retrieved January 12, 2022 from <https://skype.daesung.com/eng/main.asp>

³Mmhmm - Stunning video presentations, simply produced. Retrieved January 12, 2022 from <https://www.mmhmm.app/>

⁴Apple Facetime - FaceTime on the App Store. Retrieved January 12, 2022 from <https://apps.apple.com/us/app/facetime/id1110145091>

⁵Bunch - Group Video Chat & Party Games - Apps on Google Play. Retrieved January 12, 2022 from <https://bunch.live/>

2 PARTICIPATORY DESIGN SESSION

To explore the user needs and expectations in virtual group selfies, we conduct a brainstorming session and a follow-up group interview.

2.1 Participants

We recruited four groups of participants at a college through word-of-mouth. Each group consists of three students who knew each other very well, since we asked one per group to recruit two close acquaintances. Their average age was 25.3 years (SD = 2.02), and 66.7% of them were men. Participants in each group had various majors, including cognitive neuroscience, mechanical engineering, design, AI, nuclear engineering, and electrical engineering. All participants were familiar with at least one video conferencing tools and had sufficient experience with offline group selfies.

2.2 Procedure

The participatory design session consists of three parts. First, in order to sensitize them, we asked each group to discuss in what situation they would meet friends online. Then we introduced basic features of Virfie, and asked them to imagine various scenarios of taking group selfie online. Once every participant came up with at least three scenarios, they shared preferred scenarios within the group. Each group chose the three most interesting scenarios, elaborate them with more details, and explained to other groups. After the brainstorming session, we interviewed each group focusing on what a virtual group selfie system should provide beyond the basic features of Virfie.

2.3 Results

As a result of the participatory design session, we collected 45 hand-drawn group selfie ideas and accompanying descriptions that contextualize each idea. On average, each group created 12 scenarios. We conducted a thematic analysis on them focusing on (1) what kinds of scenarios they imagined for taking virtual group selfies, (2) what kind of interaction is mentioned, and (3) how those interactions would create social bonding. As a result, we identified three major patterns of virtual group selfie scenarios as below.

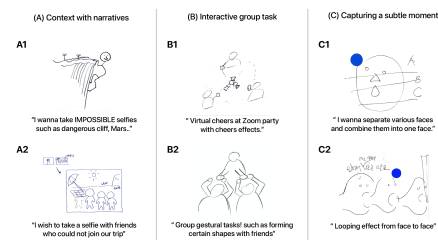


Figure 2: Proposed sketches of virtual group selfie ideas from participatory design session categorized into three concepts. (A) Context with narratives presents the participants' inherent desires to explore sites. (B) Interactive group tasks generate specific effects once the provided tasks are fulfilled by users' gestures (C) Capturing a subtle moment presents a feature of capturing users' momentary motions.

Context with narratives indicates the participants' inherent desires to meet at certain places that are difficult to access during the pandemic situation. As an example, P5 said it would be fun to take a virtual trip with a friend who could not travel together (Figure 2.A2). P1 proposed several ideas about dangerous places to take groups selfies in real world, such as posing at the edge of a high cliff with friends (Figure 2.A1) or standing above water.

Interactive group tasks represent visual effects that are triggered when users fulfill predefined tasks with their body gestures. For instance, P4 described "Cheers" effect being triggered when all the beer glasses hold by users' hands collide with each other at the center of the video frame. They also suggested a clinking sound with a firecracker-shaped animation effect to be triggered along with the "Cheers" effect (Figure 2.B1). P5 proposed another scenario where the system provides a certain shape, and users should make their bodies fit to the shape as much as possible (Figure 2.B2).

Capturing a subtle moment is a pattern of group selfies that capture user's momentary motion or subtle states. For instance, P12 proposed to split each member's face into three pieces, and continuously create new faces by randomly combining the pieces (Figure 2.C1). P10 proposed even a supernatural case of virtual group selfie, "virtual face throw up", in which each user's face is coming out from another member's mouth (Figure 2.C2).

3 VIRFIE: A VIRTUAL GROUP SELFIE STATION

3.1 Design goals

We designed Virfie with the following goals in mind.

G1. Provide a group gesture-based task to promote social interaction. Conventional video conference platforms keep individuals in their own grids, and let them indirectly interact with each other through audio and video. In contrast, we designed Virfie to let users completely overlap and interact with each other using body gestures.

G2. Support the process of focusing and capturing others. Since paying attention to non-verbal cues such facial expressions or body gestures is very costly, and most people using conventional video meetings rely on verbal information to infer others' emotions. The second design goal is to help users pay attention to others' facial expressions while taking group selfies, and to trigger a small talk and relieves the tension among users. **G3. Enhance the narratives in a virtual space with context.** While existing video conference platforms allow each user to customize his/her own background, people in a meeting cannot share the context. The third goal is to support a shared virtual space, which includes not only backgrounds but also scale, position, rotation, and z-index of individual users and images overlaid on top. We believe that achieving this goal would be able to broaden users' perspectives towards the context of virtual group selfie, and further trigger their creativity for making their own virtual spaces.

3.2 System Usage

Virfie is a web-based video communication platform that enables multiple users to take group selfies in remote circumstances. To use Virfie, a group of users visit the system's URL and join the same room. As they join the room, they can see each other's video

stream (Figure 3). Anyone in the room can choose a scenario from the dropdown menu (Figure 3.C) in which they will take group selfies. The current version of Virfie has three sample scenarios as described in section 4. Once users click "Start Video Composition" button, the main canvas (Figure 3.H) starts showing every user in the room according to the predefined scenario configuration. Virfie allows users to take snapshots of the current view at any time by pressing the camera icon at the bottom left corner of the canvas (Figure 3.I). They can also toggle the full screen mode with the button at the bottom right corner (Figure 3.J)

3.3 Three Scenarios

To demonstrate the capabilities of Virfie and to validate the design goals, we implemented three ideas chosen from the participatory design session.

1) Space Wiggle: Once users gather at the center of the screen, the vortex effect presents as if they were falling into a black hole. Both the vortex effect and the background of the universe support users' group selfie experience. Space Wiggle is designed to demonstrate the effectiveness of the first goal (G1) in Section 3.1.

2) Emotion Tracker: While Virfie does not recognize users' emotional states, four types of weather icons randomly appear above each user's face with three-seconds intervals. Emotion Tracker is designed to demonstrate the effectiveness of the second goal (G2).

3) Club Virfie: In this scenario, participants experience three scenes of a virtual club: (a) Club Entrance, (b) Club Bar, and (c) Dance Hall, illustrated with three screenshots in Figure 1C. At the Club Entrance, while users are gathering at one point, Virfie keeps taking snapshots with a polaroid-shaped frame following them. We expect this feature would trigger informal and embodied communication between users such as where to gather and how to take snapshots. In the second scene, Club Bar, Virfie tracks users' hands and puts beer glasses on them. Amid the collision of all the glasses, a visual effect of firecracker appears, reminding them of the memories of cheering in real. The last scene, Dance Hall presents the background of the crowd dancing behind users. It also displays animated lighting effects in various colors over users. This scenario is designed to demonstrate the effectiveness of the third goal (G3).

3.4 Scenario Configuration

While the three sample scenarios in Section 3.2 demonstrates the capability of Virfie, users may want to modify an existing scenario or even create a new scenario from scratch. To support the needs for custom scenarios, Virfie provides an editing panel (Figure 3F). A scenario is defined as a JSON text (i.e., key-value pairs organized in a hierarchical structure). As illustrated in Figure 4, The scenes property contains a single or multiple scenes in the current scenario. Each scene contains title, audio, players, triggers, overlays, and backgrounds parameters as detailed in Figure 4. As a user modifies the configuration, the composition view gets updated in real time. If the user presses the Update Video Composition button, all the users will have the same configuration.

Virfie's video composition has three basic layers. The bottom most layer contains a single or multiple background images. If multiple image files were set in the background property, Virfie animates those images according to the given interval parameter.

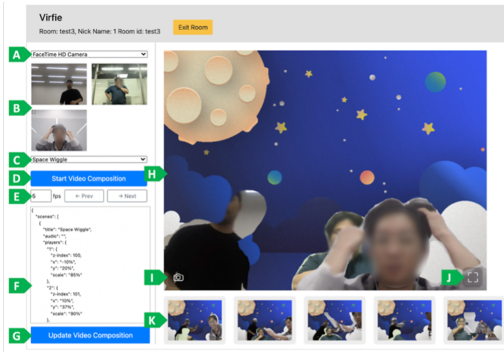


Figure 3: The user interface of Virfie consists of (A) webcam selector, (B) video streams from the users in the current room, (C) scenario selector, (D) "Start Video Composition" button, (E) refresh rate controller, (F) panel for editing scenario configuration, (G) "Update Video Composition" button, (H) main canvas for showing composed video stream, (I) button for taking snapshots, (J) button for toggling full-screen view, and (K) a list of snapshots taken so far.

In the middle layer, video streams from users are drawn over the background image. For each video stream, Virfie automatically removes its background using the TensorFlowJS's BodyPix module⁶ and draws the user only. The players parameter defines x and y coordinates, scale, and z-index of each player's video. Lastly, overlay images are drawn on top of the canvas. Same as the background, an overlay is either a single image or multiple images animated with a given interval. In addition, each overlay can be placed either at a fixed coordinate, or tracking a specific player's body position (e.g., beer glasses tracking left/right wrists of the players in the second scene of the Club Virfie scenario)

Virfie supports event handlings. A trigger parameter has two properties: the conditions property defines in what situation the event is triggered, and the actions property defines how the main canvas (Figure 3.H) updates. For example, in the Space Wiggle scenario, when all the users are at the center area, an image distortion filter (e.g., swirl effect) is applied to the canvas. In the Emotion Tracker scenario, without any condition (i.e., always) one of the four weather icons is attached on top of each user's head position. In the second scene of the Club Virfie scenario, if all the users had their left / right wrists (i.e., beer glasses) in the center, an animation sequence (i.e., bubbles splashing from the beer glasses) is played.

3.5 Implementation

The Virfie system is developed as a web application running on Chrome or other kinds of Webkit-based browsers. In order to capture input video streams via WebRTC⁷ every user's computer must enable at least one camera input. Users in the same scenario are directly connected with each other via Web Socket⁸ so that every user has an identical set of input video streams without overloading the Virfie server. On the client side, body parts are extracted from

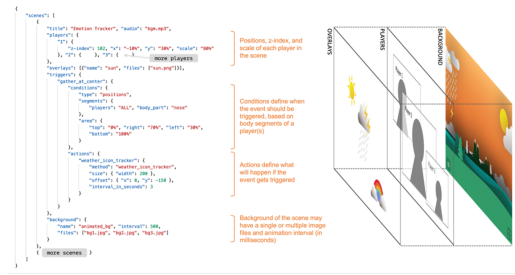


Figure 4: (left) A typical scenario configuration of Virfie and (right) how component layers are drawn to the main canvas. Each scene contains a number of properties including title, audio, players, overlays, triggers, and background. Among those properties, background, players, and overlays are sequentially drawn onto the main canvas.

every video stream in real-time using the TensorFlowJS BodyPix module.

4 USER STUDY

To validate whether Virfie supports participants' needs and the scenarios they proposed, we conducted a user study with the same participants who did the participatory design session. They took group selfies in the three sample scenarios, and reported usability issues, missing features, and further opportunities.

4.1 Procedure

To simulate the feeling of physical remoteness, we separated participants into different rooms. The user study consists of two parts. In the first part, we briefly introduce Virfie's capability and usages, and let them freely take group selfies using the laptop we provided. After experiencing the three scenarios we conducted a group interview to investigate their experience of using Virfie, to identify usability issues and needs for further development, and future applications.

4.2 Result

As results of the user study, we collected 480 virtual group selfie photos, video data of individual participants, and the interview data regarding group selfies participants took. On average each participant took 40 group selfies. Also, we provide 20 minutes for group selfie activity.

5 FINDINGS

In addition to focusing on analysing the proposed ideas from the participatory design, we also investigate users' feedbacks on the three sample scenarios. Through open coding we analysed the user study result, which includes notes of participants' behaviours, summarized transcripts of the main session, and the focus group interview. Findings of the open coding include elements that promoted remote togetherness, and design considerations for enhancing virtual group selfie experience.

⁶<https://github.com/tensorflow/tfjs-models/tree/master/body-pix>

⁷<https://webrtc.org/>

⁸https://developer.mozilla.org/en-US/docs/Web/API/WebSockets_API

5.1 Virtual environment as an instrument for creative expression

Throughout the user study session most participants actively interacted with the virtual environments. For example, P1 mentioned that she wanted to step on one of the graphic assets of the background by saying, “Oh, I want to be on Mars.” Most participants were enthusiastic about the possibility of customizing the group pose using the editing panel, as P2 said “Regular selfies allow me to change background images, but I have to keep their obvious shooting angles. However, Virfie has unlimited potential to change the background and the camera position.”

Seven participants commented on benefits of the virtual environment as a tool for customizing multimodal effects. For instance, participants told us that the background music of the Club Virfie scenario was helpful for them to be engaged in the context as P5 said, “I can’t hear you well, I’m in a club right now! Please speak louder!” Also, P3 said “It would be nicer to have 360-degree Dolby sound that moves voices along with the camera’s movements.” Such diverse nature of the virtual environment motivated the participants to make a number of creative group postures and ideas for customizing the group selfie scenarios as we observed many participants expressed what they wanted and freely explored the system’s functionalities.

5.2 Remembering Dynamics of Emotion

We identified two common motivations for recording emotions with Virfie. First, participants wanted to quickly recognize current emotional states of other members. For instance, P8 said, “With general video meetings, I usually don’t bother to check others’ facial expressions. However the weather icons make me pay more attention to their current emotion.” Second, we observed that participants enjoyed capturing short-lasting feelings with specific experiences that stimulate the feelings. P4 mentioned that increased interest in other group members’ feelings also led to more personal conversation topics that support the icebreaking: “Hey, did something bad happen? You don’t look so happy (sarcastic laugh)”

5.3 Beyond the grid of video conferencing

The vast majority of the data on Virfie was related to experiences at the moment of users interacting altogether in a virtual space. We found out that eliminating traditional boxes and gathering people affects participants’ group selfie activity in two distinct ways: **1) Alleviating Gaze Misalignment:** Participants are able to see all the group members together. Looking at all at once gives a sense of being in a virtual space of a large group rather than isolated boxes. As P8 explained, “Because everyone was able to see all of them with the same view, it was more natural and it brought me the feeling of togetherness.” One participant mentioned that Virfie reminded him of his recent virtual birthday party. P10 said, “It feels so natural that everyone in the group share the same point of view.” This finding resonates with the Together mode in Microsoft Teams [7], which makes a virtual meeting more natural and relaxing by giving participants the feeling of looking into a big virtual mirror. **2) Enhancing Remote Togetherness:** The feeling of being physically in one space influenced the participants’ togetherness. P4 argued that the consistency of space and activities would increase

the level of intimacy by referring his own experience, “As we became friends while doing the same activities in the same space, staying in the same virtual space helped us get closer. In that sense, it was good to break down physical barriers in places.” Even without any interaction with other users, P7 felt a social bond in the virtual environment where images of their body overlap, “While staying together in the weather scenario, we shoot together with matching our shoulder levels. I think that was to start feeling the bond among us.” Our findings resonate with prior research that enhanced remote togetherness with tangible user interface systems [4, 8, 10]. However, Virfie prototype shows the phenomena of improving participants’ feelings of distant connectedness without employing tangible UIs or modifying physical entities, just by changing on-screen representations of people. We believe this simple yet effective technique would contribute to development of a wide range of communication tools for enhancing remote togetherness.

5.4 Reflections on current video conferencing tools

Participants most often compared Virfie with a few video conferencing tools they regularly use such as Zoom. For instance P1 said, “Even when I meet and play online with close friends, because it’s the same environment as when I work. It leads me to experience zoom fatigue and I hate this familiar environment. However, while using the Virfie, I didn’t think it was like a working environment. It has the potential to be a crucial function to distinguish formal and informal interactions online.” Similarly, two other participants commented on the benefits of gestures as a fun element to sustain a positive online selfie experience that other video conferencing tools do not support.

6 LIMITATIONS AND FUTURE WORK

While Virfie allows users to modify the scenario configuration in real time with the Edit panel (Figure 3.F), participants found it unintuitive and cumbersome to adjust their positions and scales. Instead of modifying the scenes configuration, they physically moved around within the viewpoints of the cameras. To make it easier, next versions of Virfie should provide more intuitive methods such as drag-and-drop (for moving people) or pinch (for making them bigger / smaller). We also observed participants often tried to come front or behind the others by physically moving toward or away from the screen. Thus next versions of Virfie should automatically adjust z-depth according to such movements. Lastly, voice or gesture-based commands might be very useful especially when users want to take snapshots without moving their bodies.

7 CONCLUSION

We explored a wide range of scenarios of virtual group selfie experience through the participatory design session, and identified three characteristics of group selfies: Context with narratives, Interactive group task, and Capturing a subtle moment. In order to demonstrate the validity of the ideas from the participatory session and to examine design opportunities, we developed Virfie, a web-based prototype of virtual group selfie system. Through the user study session we found the system is effective at triggering

informal conversations, alleviating gaze misalignments, and making virtual meetings more creative and expressive. We believe our findings are informative to researchers and developers who design video conferencing systems for facilitating social bonds in remote situations.

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